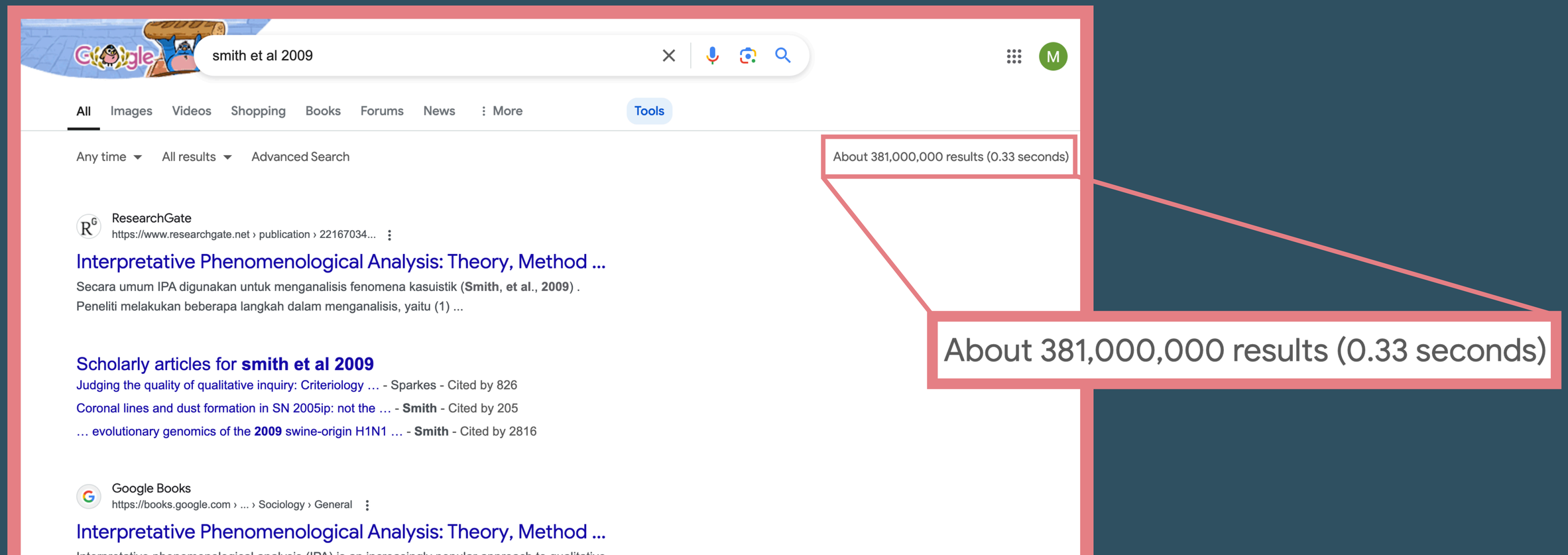
# Finding and Reading Scientific Papers

TIMESTEP Friday Workshop

2026-09-25

# Your advisor:

"Please read through Smith et al. 2009. Let me know if you have any questions. See you next week!"



# I. Where to Find a Paper




arXiv.org



NASA ADS

# Use arXiv.org or SciExplorer to find one of the following:

## A paper your advisor has asked you to read

OR one of:

- **M. Stone et al. (2025)**
(searching for galaxies orbiting a distant quasar)

- **B. Hsu et al. (2025)**
(modeling the observations of an interesting supernova)

- **N. Shah et al. (2026)**
(explaining a weird merger between two black holes)

# II. Types of Papers

Research Results Papers

Review Papers

Instrument/Software Papers

# II. Types of Papers

## Research Results Papers

Outlining an observational discovery, a theoretical model, or results from a simulation; the most common type of paper.

## Review Papers

## Instrument/Software(/Survey) Papers

# II. Types of Papers

## Research Results Papers

## Review Papers

A comprehensive look at the “state of the field” in a particular area of (astro)physics. Very long, but good for reference.

## Instrument/Software(/Survey) Papers

# II. Types of Papers

Research Results Papers

Review Papers

Instrument/Software(/Survey) Papers

Documenting a piece of work so other people in the field can use and reference it.

# Research Results Papers

- The Astrophysical Journal (ApJ, ApJL)
- The Astronomical Journal (AJ)
- The Planetary Science Journal (PSJ)
- Monthly Notices of the Royal Astronomical Society (MNRAS)
- Astronomy & Astrophysics (A&A)
- Research Notes of the AAS* (RNAAS)

# Review Papers

- Annual Review of Astronomy and Astrophysics (ARA&A)

# Instrument/Software Papers

- Proceedings of the SPIE (“SPIE”)
- Journal of Astronomical Telescopes, Instruments, and Software (JATIS)
- Journal of Open Source Software (JOSS)
- The Astrophysical Journal Supplement Series (ApJS)

# Goals of a Paper:

- **Provide background and motivation.** What prompted the authors to do this work? What background information is relevant to the project?
- **Explain methods.** How did the authors do the project? What procedures did they follow? What tools did they use?
- **Outline results and their importance.** What has come out of this work? (advancing understanding in the field/providing a useful resource/deploying a new technology/allowing for a new type of observation)
- **Supply "take-home message."** What conclusions do the authors draw from the project? Do these results prompt any follow-up questions or future work?

| | | |
|---|---|---|
| Initial summary | → | Abstract |
| Provide background and motivation | → | Introduction |
| Explain methods | → | Data/Analysis/Methods/ Observations/Procedure |
| Outline results and their importance | → | Results/Discussion/ Interpretation/? |
| Supply “take-home message” | → | Conclusions |

# What section headings does your paper have?

Abstract

Introduction

Methods

Results

Conclusions

Not all of these are equally important!

# ABSTRACT:

A short (~250 word) summary of the entire paper

Contains brief summaries of future sections: introduction, methods, and results

"What will I find if I read this paper?"

## Measuring Star Formation and Black Hole Accretion Rates in Tandem Using Mid-infrared Spectra of Local Infrared Luminous Galaxies

Meredith Stone[1], Alexandra Pope[1], Jed McKinney[1], Lee Armus[2], Tanio Díaz-Santos[3], Hanae Inami[4], Allison Kirkpatrick[5], and Sabrina Stierwalt[6]

[1] Department of Astronomy, University of Massachusetts, Amherst, MA 01003, USA
[2] Spitzer Science Center, California Institute of Technology, Pasadena, CA 91125, USA
[3] Institute of Astrophysics, Foundation for Research and Technology-Hellas (FORTH), Heraklion, GR-70013, Greece
[4] Hiroshima Astrophysical Science Center, Hiroshima University, 1-3-1 Kagamiyama, Higashi-Hiroshima, Hiroshima 739-8526, Japan
[5] Department of Physics and Astronomy, University of Kansas, Lawrence, KS 66045, USA
[6] Physics Department, Occidental College, 1600 Campus Road, Los Angeles, CA 90041 USA

*Received 2021 December 14; revised 2022 April 15; accepted 2022 June 8; published 2022 July 21*

**Abstract**

We present the results of a stacking analysis performed on Spitzer/Infrared Spectrograph high-resolution mid-infrared (mid-IR) spectra of luminous infrared galaxies (LIRGs) in the Great Observatories All-Sky LIRG Survey. By binning in relation to mid-IR active galactic nucleus (AGN) fraction and stacking spectra, we detect bright emission lines [Ne II] and [Ne III], which trace star formation, and fainter emission lines [Ne V] and [O IV], which trace AGN activity, throughout the sample. We find that the [Ne II] luminosity is fairly constant across all AGN fraction bins, while the [O IV] and [Ne V] luminosities increase by over an order of magnitude. Our measured average line ratios, [Ne V]/[Ne II] and [O IV]/[Ne II], at low AGN fraction are similar to H II galaxies, while the line ratios at high AGN fraction are similar to LINERs and Seyferts. We decompose the [O IV] luminosity into star formation and AGN components by fitting the [O IV] luminosity as a function of the [Ne II] luminosity and the mid-IR AGN fraction. The [O IV] luminosity in LIRGs is dominated by star formation for mid-IR AGN fractions $\lesssim$0.3. With the corrected [O IV] luminosity, we calculate black hole accretion rates (BHARs) ranging from $10^{-5}$ $M_\odot$ yr$^{-1}$ at low AGN fractions to 0.2 $M_\odot$ yr$^{-1}$ at the highest AGN fractions. We find that using the [O IV] luminosity, without correcting for star formation, can lead to overestimation of the BHAR by up to a factor of 30 in starburst-dominated LIRGs. Finally, we show that the BHAR/star formation rate ratio increases by more than three orders of magnitude as a function of mid-IR AGN fraction in LIRGs.

*Unified Astronomy Thesaurus concepts:* Luminous infrared galaxies (946); Ultraluminous infrared galaxies (1735); Supermassive black holes (1663); Star formation (1569); Galaxy evolution (594)

*Supporting material:* data behind figure

what we did (methods)
what we found (results)

# FIGURES (& tables):

Most critical parts of the paper

Check caption: what are the axes showing? What do the colors mean?

Think back to the abstract: how does the figure fit in to the entire paper?

Write down your questions!

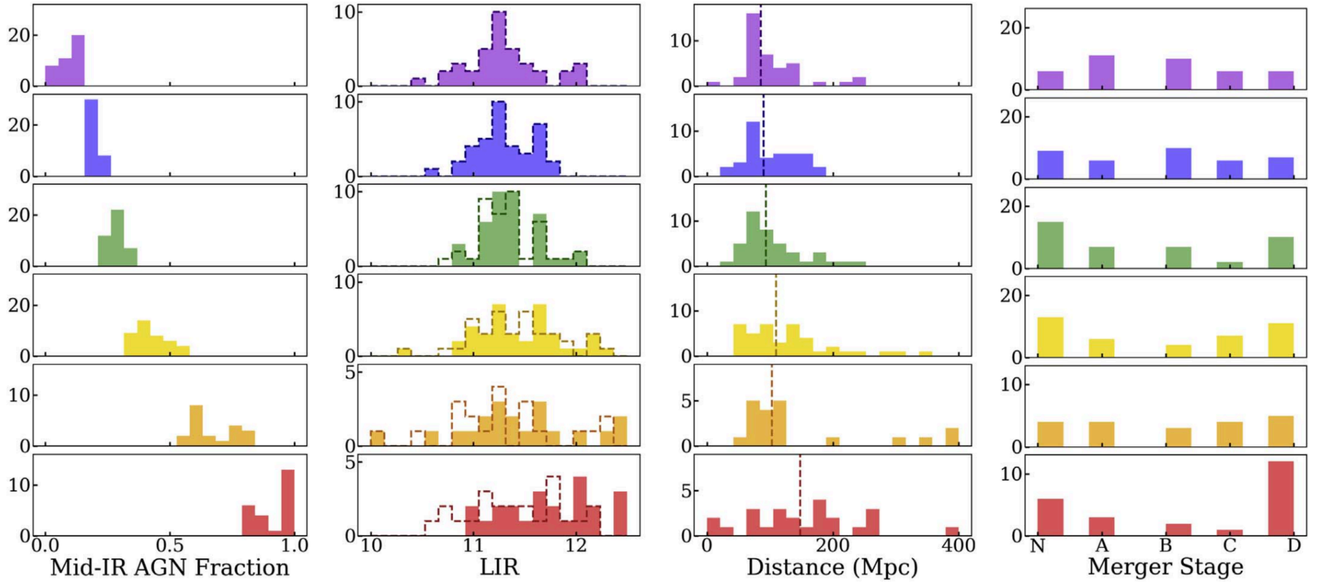



**Figure 1.** Distributions of the mid-IR AGN fraction (Díaz-Santos et al. 2017) from the 6.2 $\mu$m PAH EQW (Stierwalt et al. 2013), the IR luminosity (Díaz-Santos et al. 2017), the luminosity distance (Armus et al. 2009), and the merger stage (Stierwalt et al. 2013) for our six AGN fraction bins. We use different colors to denote the six bins of AGN fraction, from purple, at the lowest AGN fractions, to red, at the highest AGN fractions. In the second column, the total IR luminosity, $L_{IR}$, is the solid distribution, while the dashed distribution is just the IR luminosity from star formation, $L_{IR,SF}$. The vertical lines in the third column show the median distance for each bin. The merger stages from Stierwalt et al. (2013) are N (nonmergers), A (premergers), B (early-stage mergers), C (mid-stage mergers), and D (late-stage mergers).

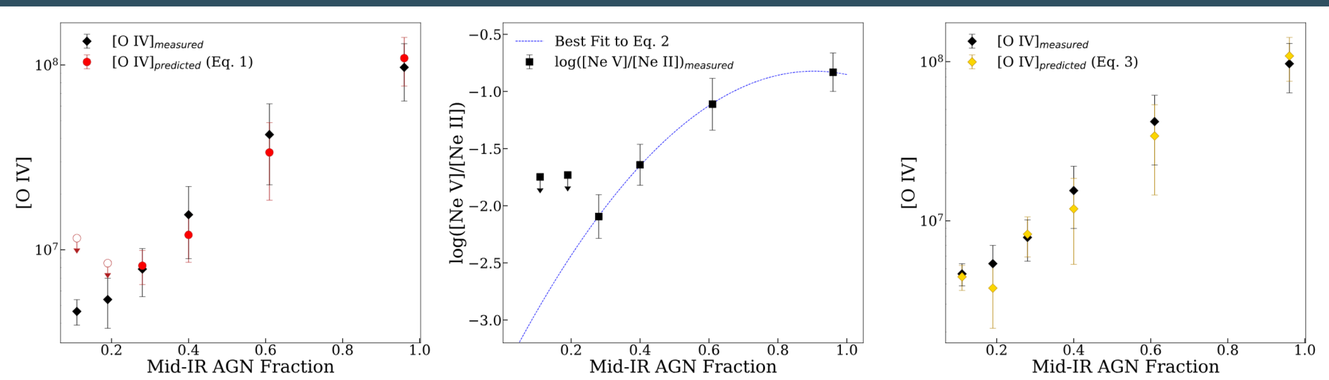



**Figure 10.** The process of predicting the [O IV] luminosity from the [Ne II] luminosity and AGN fraction. We fit [O IV] as a function of [Ne II] and [Ne V] (left panel and Equation (1)), and fit log([Ne V]/[Ne II]) as a function of AGN fraction (center panel and Equation (2)). We then combine these two relations to predict [O IV] as a function of [Ne II] luminosity and AGN fraction (right panel). Our relation in terms of [Ne II] and AGN fraction reproduces the [O IV] measurements, including in Bins 1 and 2, where [Ne V] is not detected. All [Ne II] fluxes in this figure are unscaled.

# What question(s) do you have from your paper’s figures?

# CONCLUSIONS:

Typically a more fleshed-out version of the abstract

What did they do? What results did they obtain? Why does it matter?

Results will be summarized in more detail than in the abstract

Can you match the figures you saw to the conclusions?

## 6. Summary and Future Applications

We perform a spectral stacking analysis as a function of mid-IR AGN fraction using 203 Spitzer/IRS spectra of the GOALS sample in order to explore the coevolution of the mid-IR atomic emission lines 12.8 $\mu$m [Ne II], 14.3 $\mu$m [Ne V], 15.5 $\mu$m [Ne III], and 25.9 $\mu$m [O IV].

1. We detect all lines in all bins, except for [Ne V] in our two lowest–AGN fraction stacked spectra, where we derive 3$\sigma$ upper limits.
2. We compare the line ratios to results from the literature for other samples of galaxies, confirming that our low–AGN fraction stacked samples are consistent with H II galaxies and that our high–AGN fraction bins most closely resemble LINER or Seyfert 2 galaxies.
3. We derive a relationship between the [Ne II] luminosity, the AGN fraction, and the [O IV] luminosity, allowing us to determine the relative contributions to [O IV] from star formation and the AGN. This relation can be used to correct for the contamination of [O IV] due to star formation, given the AGN fraction.
4. We apply this empirical relation to predict the [O IV] luminosities of individual galaxies in the GOALS samples, and find excellent agreement with the measurements.
5. Using our [O IV] decomposition, we calculate BHARs for stacked spectra from $[O\,IV]_{AGN}$, and quantify the evolution of the BHAR/SFR ratio as a function of AGN fraction. This ratio spans over three orders of magnitude across the GOALS sample.

We have used the technique of spectral stacking analysis to detect faint star formation signals in AGN-dominated sources and faint AGN signals in star formation–dominated sources. We intend to apply this technique to study additional spectral lines in the low- and high-resolution spectra, including PAH lines, molecular hydrogen ($H_2$) rotational lines, and fainter atomic lines, such as [Cl II] and [Fe II]. The differences in the distributions of the merger stages between our bins motivate the extension of our stacking analysis to examine the evolution of mid-IR spectral features as a function of merger stage.

The James Webb Space Telescope (JWST) will provide renewed access to the mid-IR spectral regime at unprecedented spectral resolution and sensitivity. Our stacking analysis provides average mid-IR line fluxes for IR-selected galaxies as a function of AGN fraction, which will be helpful for planning JWST observations. Our empirical formula could be useful for estimating [O IV] fluxes for future studies of LIRGs when only the [Ne II] and mid-IR AGN fractions are known. In addition to the key star formation and AGN line measurements presented in this paper, we make our stacked spectra publicly available.[10] With the JWST/MIRI integral field unit, mid-IR lines can be spatially resolved in individual nearby galaxies, with [Ne V] and [Ne II] both observable in galaxies out to $z \lesssim 1$. Furthermore, the spectral resolution of JWST can be exploited to study the kinematics of the gas (Inami et al. 2013). The physical extent and velocity of the star formation and AGN line emission will provide further diagnostics on the coevolution of these processes in galaxies. Over the longer term, a new cold far-IR space telescope will be required to probe these powerful diagnostics in galaxies over all cosmic time, to map the relative buildup of stars and supermassive black holes.

Summarizing the main points of the paper

Discussing future research

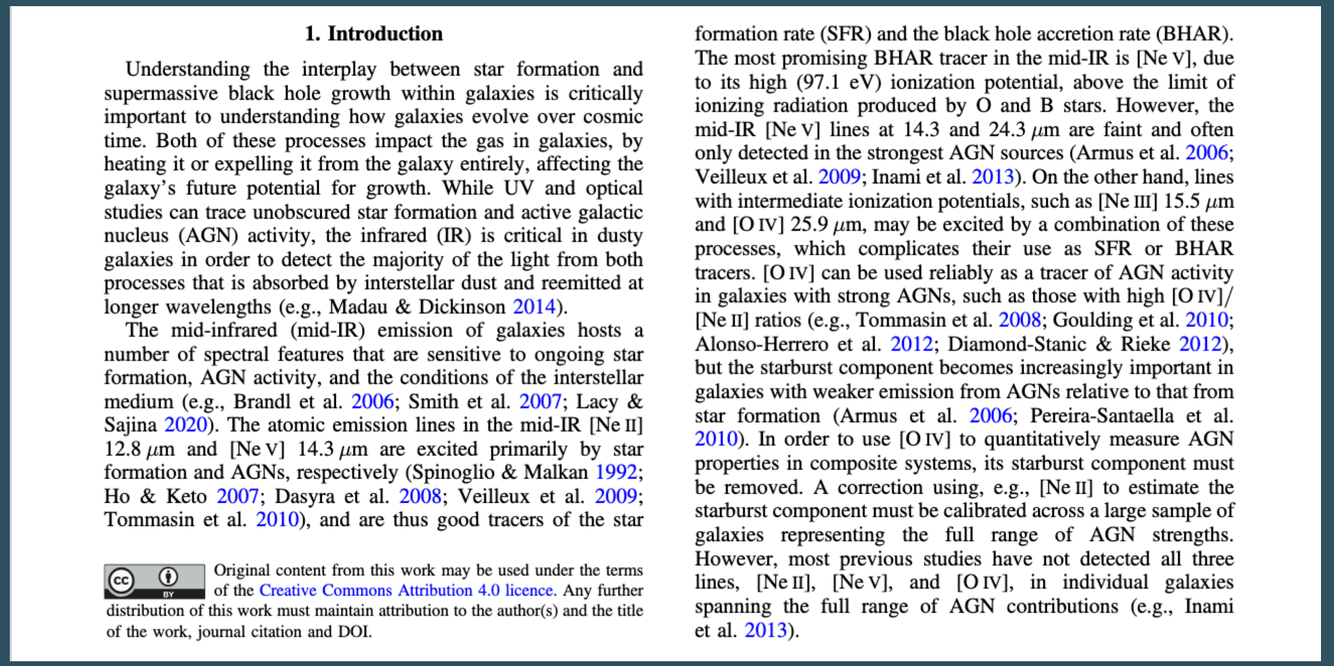
stop here, or…

# INTRODUCTION:

High-level background and motivation for the research

More of a “reminder” of concepts than true “intro”: assumes prior knowledge

In-text citations link to other important papers in the field



## 1. Introduction

Understanding the interplay between star formation and supermassive black hole growth within galaxies is critically important to understanding how galaxies evolve over cosmic time. Both of these processes impact the gas in galaxies, by heating it or expelling it from the galaxy entirely, affecting the galaxy’s future potential for growth. While UV and optical studies can trace unobscured star formation and active galactic nucleus (AGN) activity, the infrared (IR) is critical in dusty galaxies in order to detect the majority of the light from both processes that is absorbed by interstellar dust and reemitted at longer wavelengths (e.g., Madau & Dickinson 2014).

The mid-infrared (mid-IR) emission of galaxies hosts a number of spectral features that are sensitive to ongoing star formation, AGN activity, and the conditions of the interstellar medium (e.g., Brandl et al. 2006; Smith et al. 2007; Lacy & Sajina 2020). The atomic emission lines in the mid-IR [Ne II] 12.8 $\mu$m and [Ne V] 14.3 $\mu$m are excited primarily by star formation and AGNs, respectively (Spinoglio & Malkan 1992; Ho & Keto 2007; Dasyra et al. 2008; Veilleux et al. 2009; Tommasin et al. 2010), and are thus good tracers of the star formation rate (SFR) and the black hole accretion rate (BHAR). The most promising BHAR tracer in the mid-IR is [Ne V], due to its high (97.1 eV) ionization potential, above the limit of ionizing radiation produced by O and B stars. However, the mid-IR [Ne V] lines at 14.3 and 24.3 $\mu$m are faint and often only detected in the strongest AGN sources (Armus et al. 2006; Veilleux et al. 2009; Inami et al. 2013). On the other hand, lines with intermediate ionization potentials, such as [Ne III] 15.5 $\mu$m and [O IV] 25.9 $\mu$m, may be excited by a combination of these processes, which complicates their use as SFR or BHAR tracers. [O IV] can be used reliably as a tracer of AGN activity in galaxies with strong AGNs, such as those with high [O IV]/[Ne II] ratios (e.g., Tommasin et al. 2008; Goulding et al. 2010; Alonso-Herrero et al. 2012; Diamond-Stanic & Rieke 2012), but the starburst component becomes increasingly important in galaxies with weaker emission from AGNs relative to that from star formation (Armus et al. 2006; Pereira-Santaella et al. 2010). In order to use [O IV] to quantitatively measure AGN properties in composite systems, its starburst component must be removed. A correction using, e.g., [Ne II] to estimate the starburst component must be calibrated across a large sample of galaxies representing the full range of AGN strengths. However, most previous studies have not detected all three lines, [Ne II], [Ne V], and [O IV], in individual galaxies spanning the full range of AGN contributions (e.g., Inami et al. 2013).

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# RESULTS/DISCUSSION:

Like solving a puzzle: don’t just read, look for answers. Highlight! Take notes!

What’s the context for those figures you saw?

Can you answer some of the questions that you wrote down while looking at the figures and conclusion?

Can you see the bullet points from the conclusion taking shape in this section?

stop here, or…

# METHODS
## (or another name)

Detailed, technical, jargon-heavy explanation of data

How was the telescope data reduced? What physics did the simulation include?

Any lingering questions about the *data,* rather than the results, might be answered here

Some figures, maybe

### 2. Sample and Data

Our sample consists of GOALS galaxies with high-resolution ($R = 600$) Spitzer/IRS data in both the short–high (SH) and long–high (LH) channels (Inami et al. 2013). We retrieved the 1D reduced spectra from the Infrared Science Archive (GOALS Team 2020). These spectra were extracted using a full-slit extraction, and dedicated backgrounds were observed for subtraction: further details of the data reduction can be found in Inami et al. (2013).

We excluded 10 GOALS galaxies with very noisy spectra. In addition, we required each galaxy to have a mid-IR AGN fraction calculated from the EQW of the 6.2 $\mu$m PAH line; these are tabulated in Díaz-Santos et al. (2017) and Stierwalt et al. (2013). We use this AGN fraction indicator because it measures the relative strength of the AGNs in the mid-IR (e.g., Desai et al. 2007), where the AGNs contribute significantly, and because it does not depend on the mid-IR atomic lines that we aim to measure. We remove four galaxies whose PAH EQW-implied mid-IR AGN fractions were inconsistent with other AGN indicators, mainly the mid-IR continuum slope, and thus would have added substantial noise to the stacks had they been included. Our final sample, after these 14 nuclei were removed, consisted of 203 galaxy nuclei from the GOALS survey. The luminosity distances, total IR luminosities (8–1000 $\mu$m, $L_{IR}$), and merger stages for each galaxy were retrieved from Armus et al. (2009), Díaz-Santos et al. (2017), and Stierwalt et al. (2013). Since the AGNs will contribute to $L_{IR}$, we use the relation from Kirkpatrick et al. (2015), derived from mid-IR spectra of LIRGs at $0.3 \leqslant z \leqslant 2.8$, to convert the mid-IR AGN fraction to a bolometric AGN fraction, and use this to determine the fraction of $L_{IR}$ from star formation: $L_{IR,SF}$. Bolometric AGN fractions of the GOALS sample derived following the formalism of Veilleux et al. (2009) are tabulated in Díaz-Santos et al. (2017). These agree well with the bolometric AGN fractions derived from Kirkpatrick et al. (2015) up to a mid-IR AGN fraction of $\sim$0.9 (which corresponds to a bolometric AGN fraction of $\sim$0.5), above which the Díaz-Santos et al. (2017) bolometric AGN fractions increase very steeply up to 1. As the Díaz-Santos et al. (2017) AGN fractions are based on templates of sources that more closely resemble Palomar-Green quasars, and as the GOALS sample is IR-selected, we elect to use the Kirkpatrick et al. (2015) relation to derive bolometric AGN fractions and $L_{IR,SF}$, but note that our general results and the trends in the data are not strongly affected by the small number of sources with the highest AGN fractions.

### 3. Analysis

In this work, we study the relationships between the luminosities of 12.8 $\mu$m [Ne II], 14.3 $\mu$m [Ne V], 15.5 $\mu$m [Ne III], and 25.9 $\mu$m [O IV] as a function of mid-IR AGN fraction.

#### *3.1. AGN Fraction Bins*

For the analysis in this paper, we split our 203 galaxies into six bins of mid-IR AGN fraction. We choose to use the mid-IR AGN fraction derived from the EQW of the 6.2 $\mu$m PAH feature, used in other studies as a binning characteristic of IRS spectra (e.g., Lambrides et al. 2019), so our AGN fractions are independent of the atomic features we analyze in this paper. We aim to evenly span the range of AGN fractions from 0–1 with roughly equal number of galaxies per bin. Due to lower

# REFERENCES:

Alphabetical list of the other papers cited in the text

Good place to start if you're looking for further reading: look on SciX!


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[10] https://people.astro.umass.edu/~pope/Stone2022/



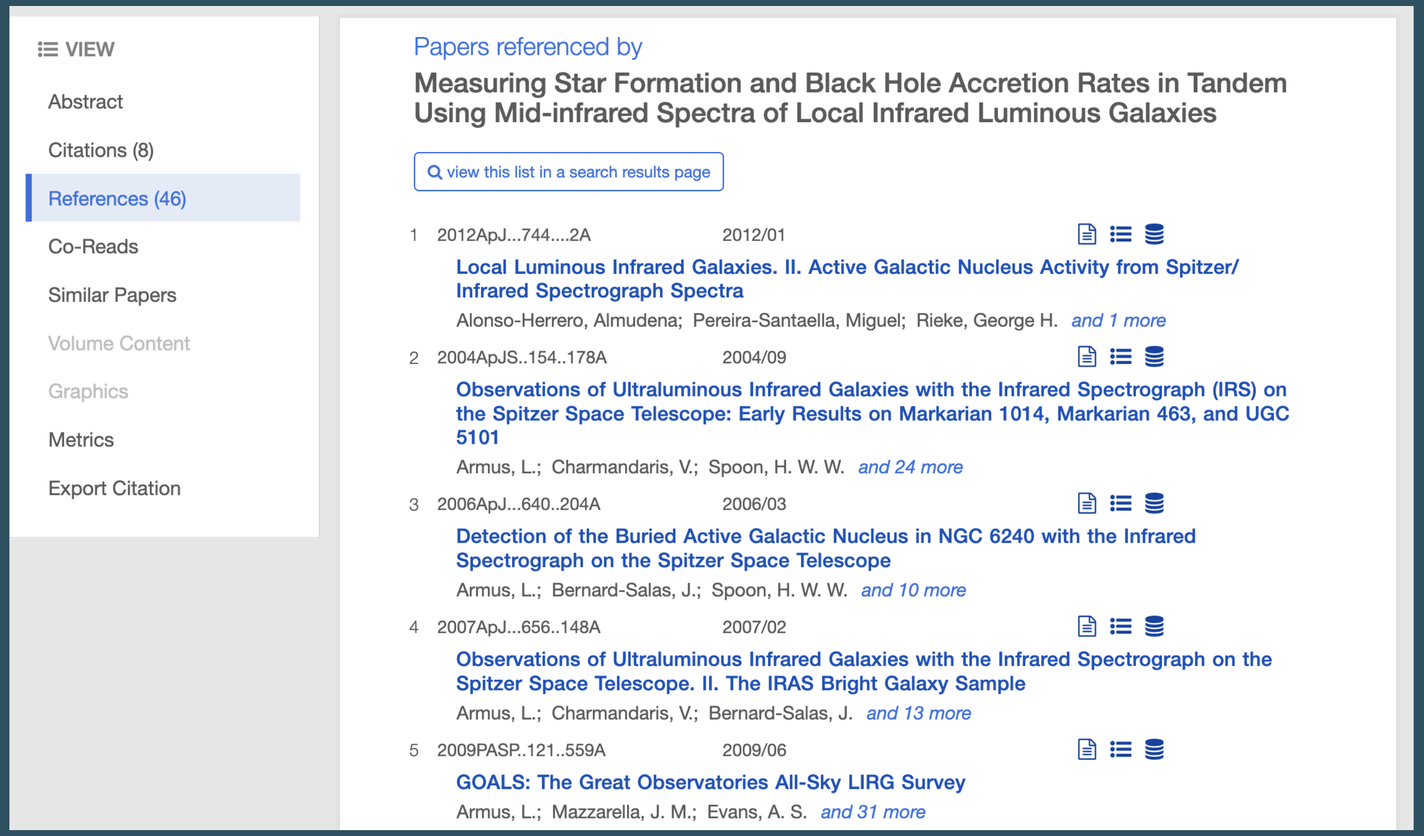


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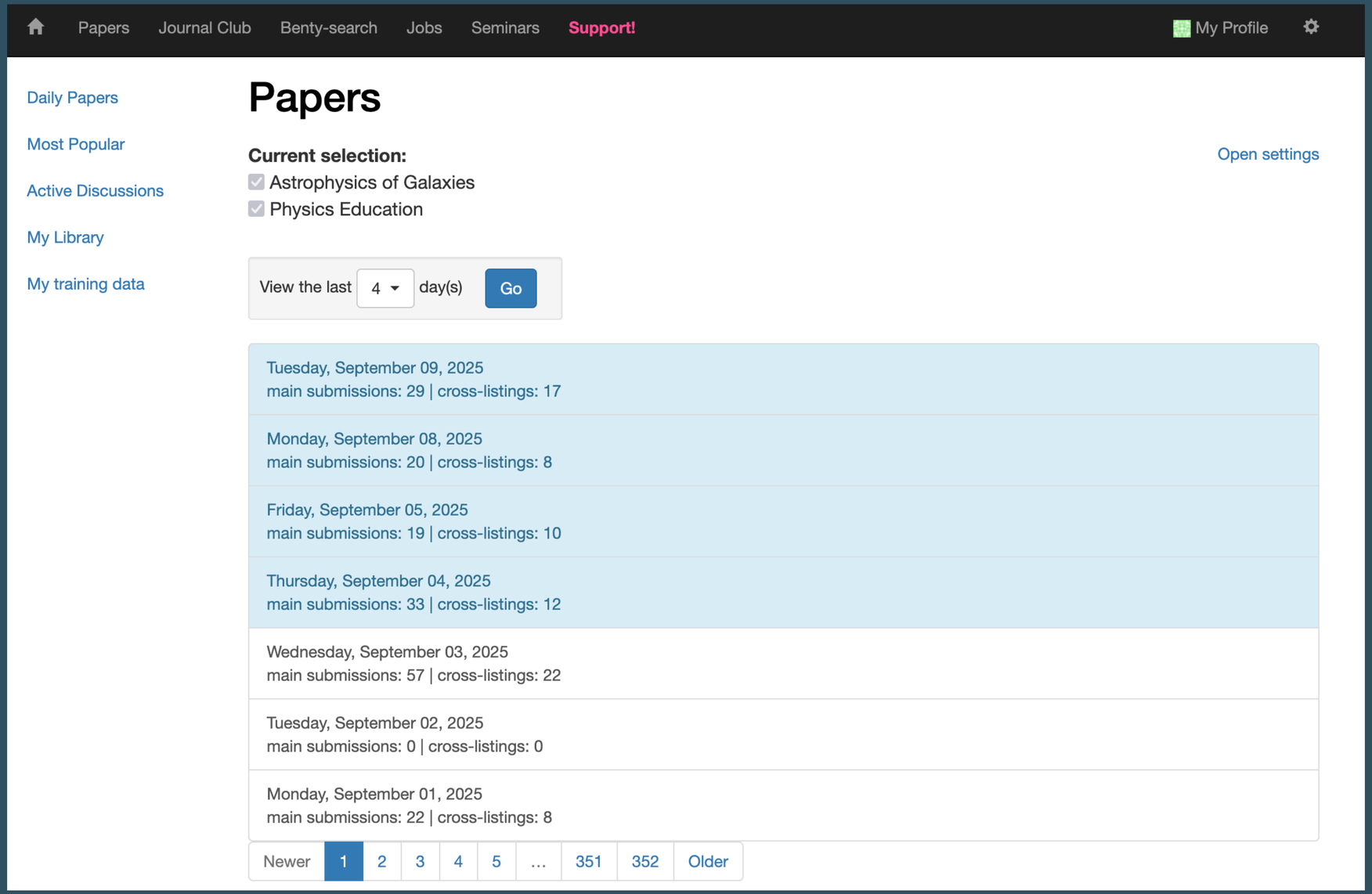
# REFERENCES:

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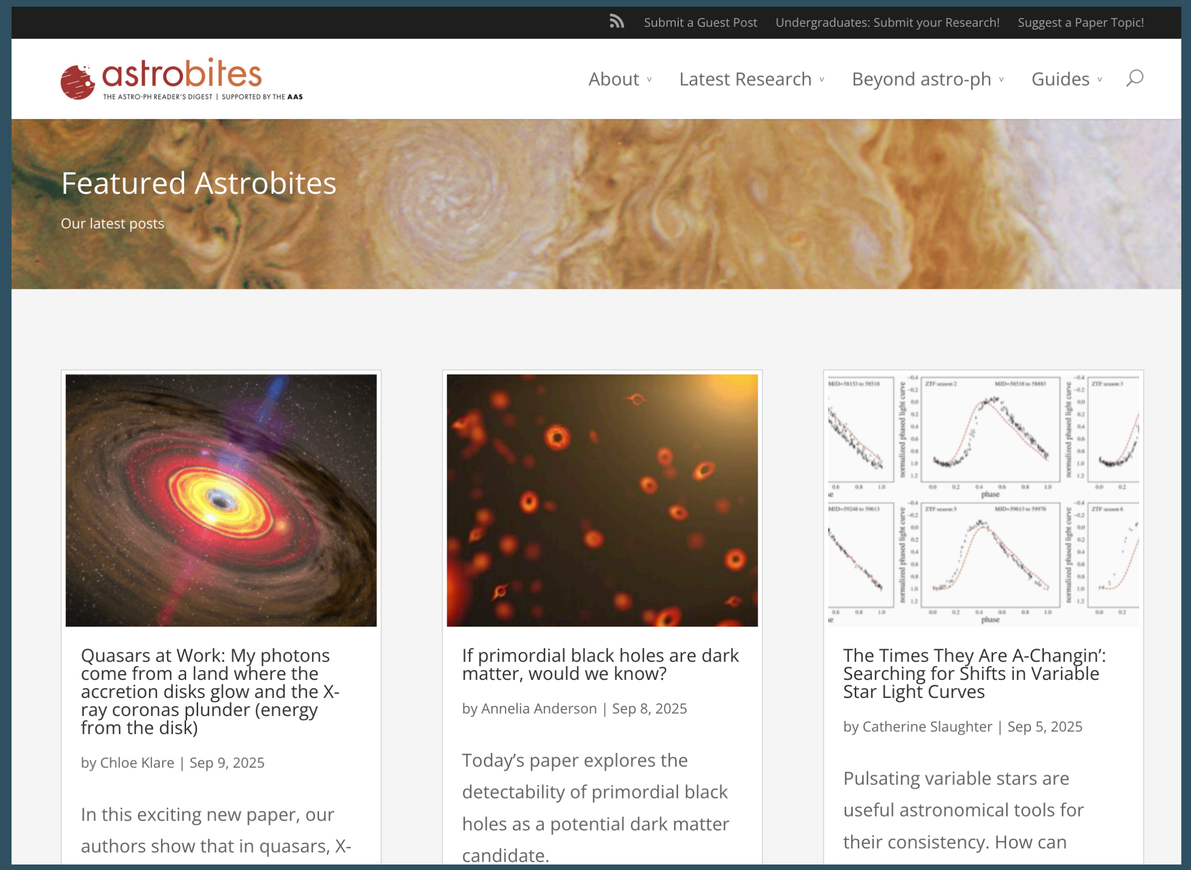
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# Find a paper in the references that seems interesting or relevant to your project.

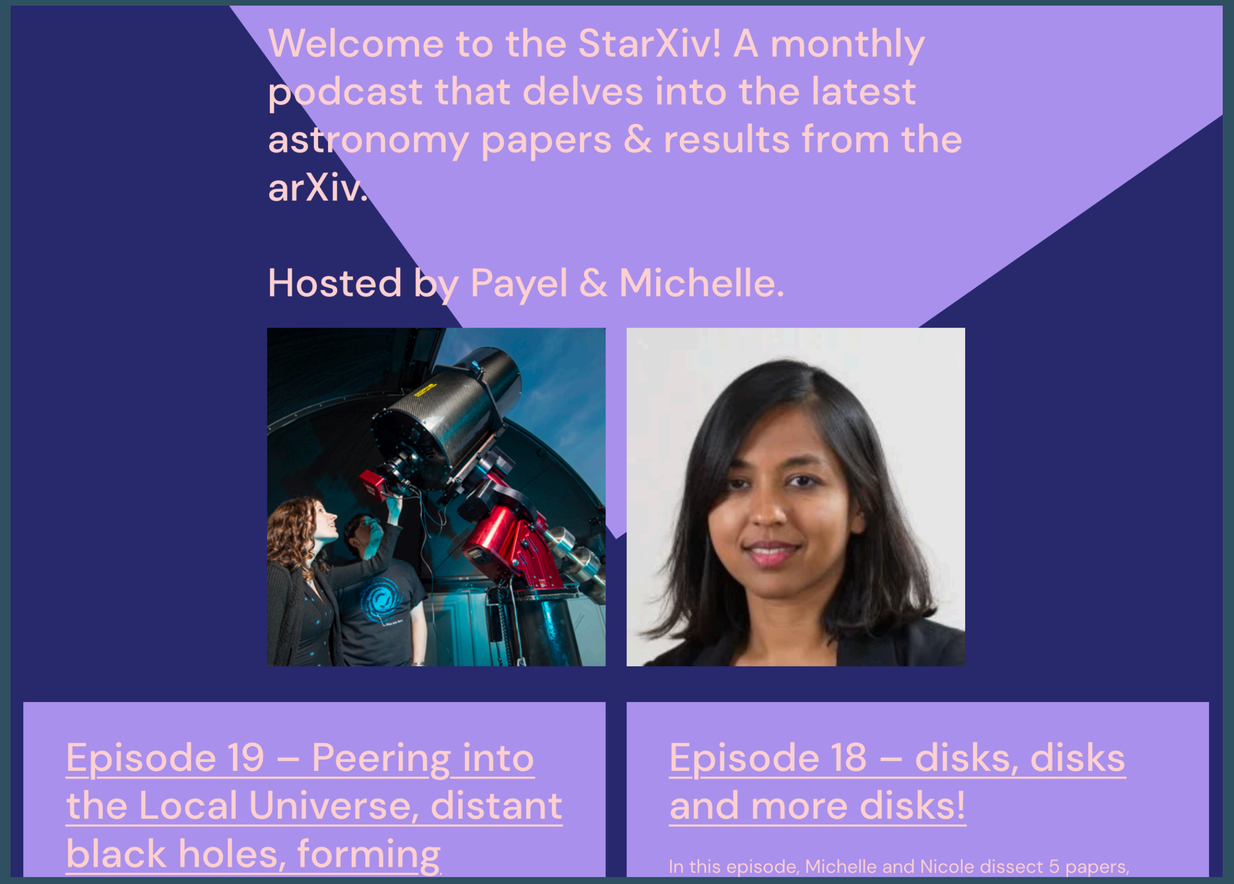



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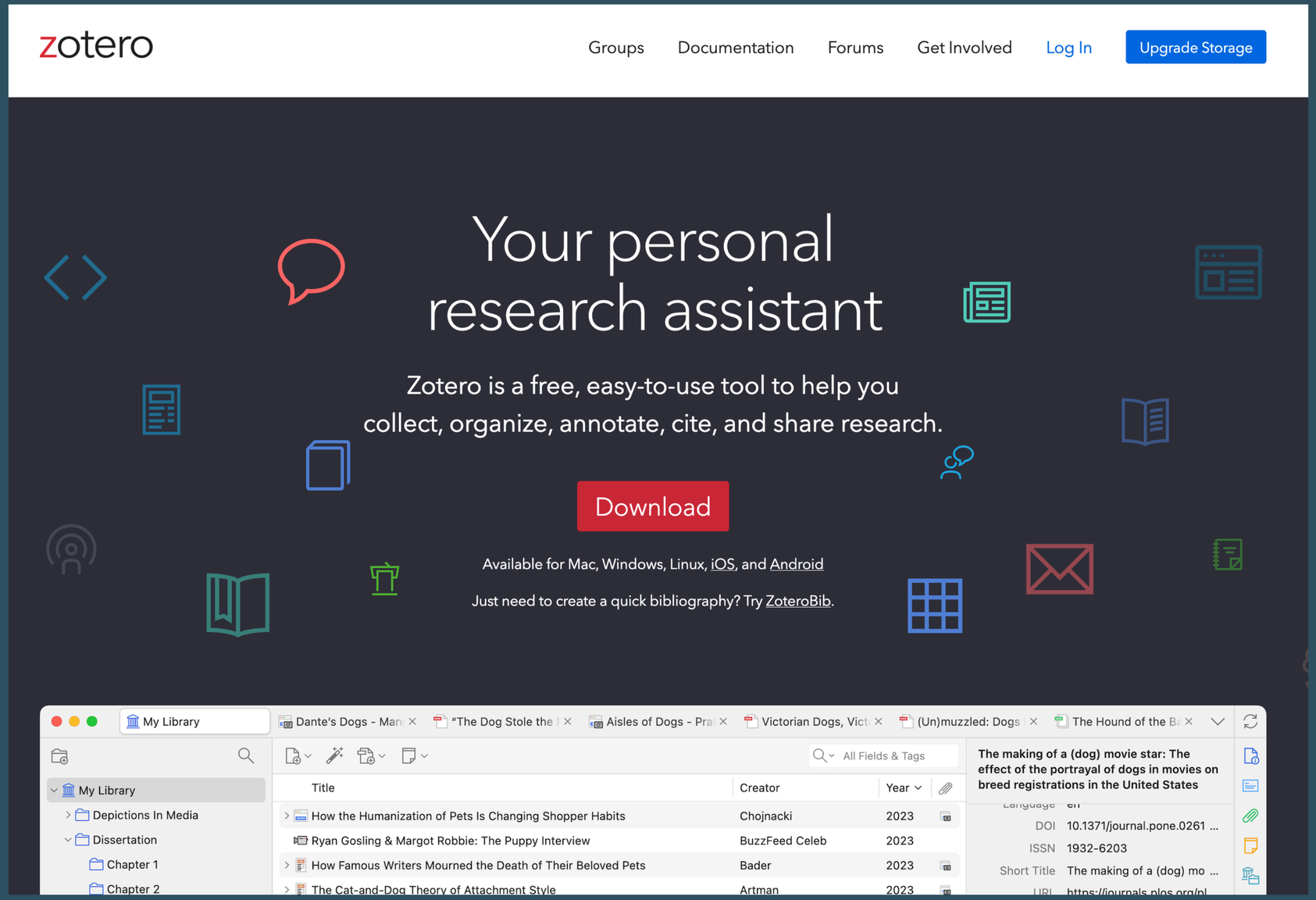
New summaries of astronomy papers every day, specifically for undergrads!




**StarXiv podcast**

Monthly discussion of interesting astro-ph papers

# “How do I keep track of all these papers?”



# Monthly Research Updates

- **Get used to presenting.** Can you explain what you're doing to a newbie? Can you summarize a month's work in only a couple of minutes?
- **Learn what other people are doing.** "Oh, they're also studying exoplanets!" "Those plots look a lot like mine."
- **Learn about research in other subfields.** What sounds interesting? What skills do you need to do that research?
- **Practice the skills you're learning.** Making plots! Writing captions! Breaking down jargon!

# The Expectation

- **Everyone will present once every four weeks.** We will remind you a week in advance.
- **Strict 5 minute time limit.** Practice summarizing!
- **Format**
  1. **Title slide**
  2. **Basic info:** Your name, your advisor, your project and **why it matters**
  3. **What you've been doing**
  4. **A challenge you've been working through**
  5. **Jargon.** What terms do you now understand that you can pass on to your peers?

# The Rotation

**Group 1 (Oct. 2nd):** Bhavya, Nathan, Ian, Stella, Seb, Maia

**Group 2 (Oct. 9th):** Jack, Maggie, Vinny, Naomi, Maurya, Dhyey

**October 16th:** Nobody (astronomy symposium)

**Group 3 (Oct. 23rd):** Fiona, Katya, Tulsi, Mohammad, Tammy, Grace

**Group 4 (Oct. 30th):** Ben, Rehman, Briscoe, Ella, Jacob, Will

# Running Models of Tidal Disruption Events

Example Student, working with Example Professor

TIMESTEP Research Update, 2026-09-25

- **Tidal Disruption Event:** when a star passes too close to a black hole and gets ripped apart; produces a bright flash of light
- Looking at the light from the TDE tells us about the properties of the black hole and the star

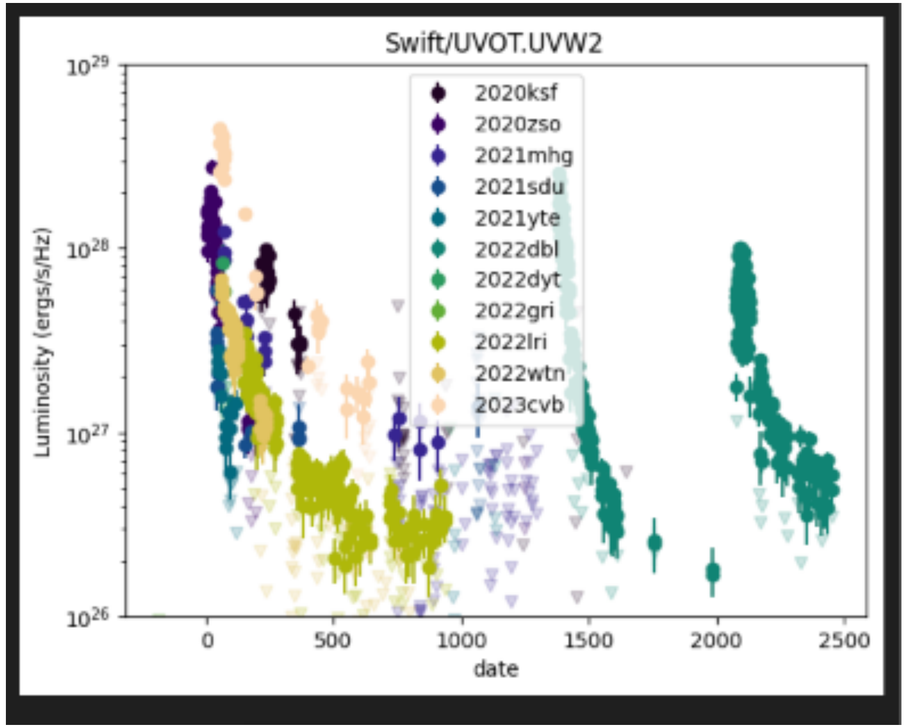
Image from STScI

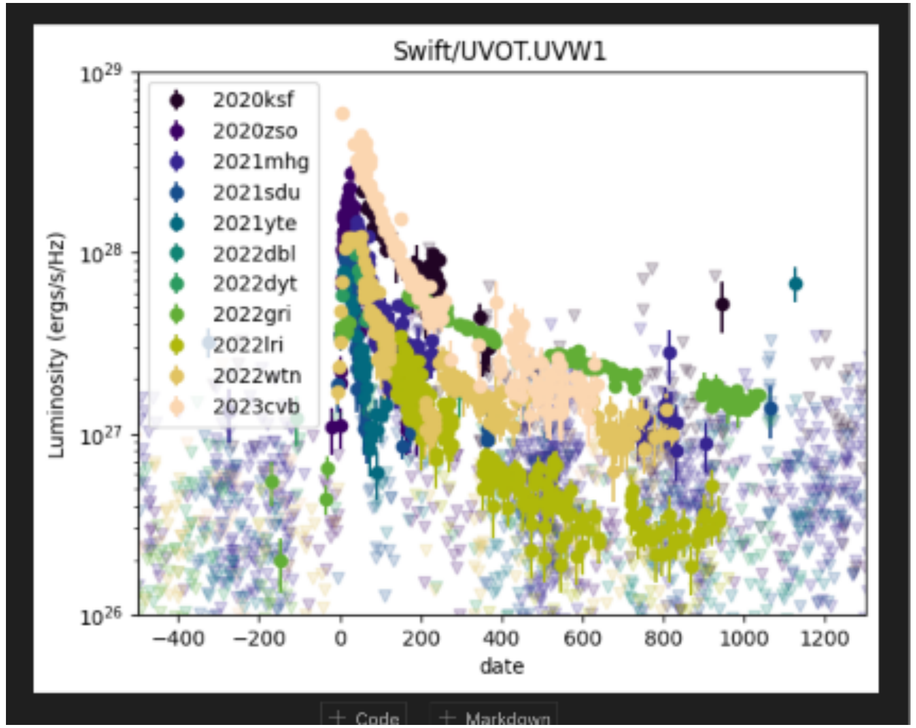


- Requires complicated modeling
- **My project:** model the same TDE data with different modeling softwares to see how the results change
- **Why it matters:** How well do the models work? Is everyone getting consistent results no matter what software they use?

- **What I'm doing:** learning how to download and plot TDE data
  - Not pretty :(

- **Challenge:** learning how to use the UA supercomputers (HPC)
  - Difficult to set up, lots of stuff you can only do through the terminal
  - Can't tell easily when it's going to crash or why



# Jargon

- **Tidal Disruption Event:** when a star passes too close to a black hole and gets ripped apart; produces a bright flash of light
- **Light Curve:** a plot of brightness over time, so you see something getting brighter or fainter
- **OTTER:** Open mulTiwavelength Transient Event Repository. A website that hosts data for lots of TDEs that you can look at and download.
- **HPC:** High-performance computing. Lots of smaller computers that are linked together into a supercomputer that can run code a lot faster than a personal computer